MethylLlama: self-supervised foundation modeling of human DNA methylation profiles

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Abstract

DNA methylation is an informative biomarker of disease, tissue identity, and biological age, but conventional task-specific models do not fully exploit context-dependent regulation across the methylome or the large volume of public profiles lacking harmonized phenotype labels. Here we present MethylLlama, a self-supervised foundation model for human DNA methylation, built on a Transformer encoder and pretrained without phenotype supervision on 169,120 profiles from diverse tissues and studies, represented at 49,156 curated CpG sites. MethylLlama encodes CpG identity, quantitative methylation state, and genomic position. We adapt Whole-Cell Expression Decoding to methylation as the Whole-CpG Encoder–Decoder (WCED), which maps independently sampled partial views of each profile to global sample representations. Each representation is trained to reconstruct CpG values excluded from its input, while an InfoNCE objective aligns representations of the same sample across views. On held-out profiles, MethylLlama preserved sample identity across partial measurements and reconstructed withheld methylation values. Frozen representations captured age- and tissue-associated variation but also substantial study-specific structure. In chronological-age prediction, a five-model ensemble outperformed matched MethylGPT and ElasticNet pipelines on a common held-out cohort. These findings support MethylLlama as a reusable methylation representation model while identifying cross-study generalization as an important remaining challenge.

1 Introduction

DNA methylation is a covalent epigenetic modification that contributes to transcriptional regulation, chromatin organization, genome stability, and the maintenance of cellular identity. At CpG dinucleotides, methylation is commonly quantified as a continuous beta value using array-based assays that profile hundreds of thousands of loci [1, 2]. The biological meaning of a methylation measurement depends on both genomic and cellular context: regulatory CpGs can reflect altered gene activity, whereas tissue-specific methylation programs capture differentiation and cell composition [3, 4]. A methylation profile therefore represents an integrated molecular state rather than a collection of independent biomarkers.

The stability of DNA methylation, its accessibility across biological specimens, and its responsiveness to development, ageing, environmental exposure, and disease have made it a valuable

substrate for biomarker discovery. Epigenetic clocks illustrate this potential. Early regularized models estimated chronological age across tissues [5], while later clocks were developed to capture mortality risk, health-related ageing, and the pace of physiological change [6–9]. These models remain strong and interpretable predictors, but each is optimized for a predefined endpoint; biological structure shared across methylation datasets must therefore be learned repeatedly for different phenotypes.

This task-specific paradigm is poorly matched to the scale and heterogeneity of public methylation data. Existing profiles span tissues, cohorts, disease states, and array platforms, while high-quality phenotype labels are available for only a subset of samples and differ among studies. Moreover, platform-defined CpG panels overlap only partially, so models must often operate on different subsets of the methylome. Conventional linear predictors perform well when the endpoint and CpG panel are fixed, but their predominantly additive formulation does not directly model context-dependent relationships among loci. Deep supervised models can capture such relationships, yet training a separate model for each application leaves the much larger unlabelled corpus unused and can generalize poorly from small labelled cohorts. These constraints motivate a foundation-model strategy that separates large-scale representation learning from task-specific supervision.

Self-supervised pretraining has enabled reusable representations in language, genomics, and single-cell transcriptomics [10–13]. In DNA methylation, MethylGPT established that a Transformer can be pretrained on a large collection of human methylomes and adapted to biological prediction tasks [14]. MethylGPT combines masked CpG prediction with whole-profile reconstruction from a global CLS representation, thereby encouraging that representation to retain sample-level information. Whole-profile decoding, however, does not explicitly require the representation of a sample to remain stable when different CpG subsets are observed, nor does it isolate recovery of loci that were entirely absent from the encoder input. These properties are important when a pretrained representation must transfer across incomplete profiles and non-identical measurement panels.

Here we present MethylLlama, a self-supervised foundation model for human DNA methylation. MethylLlama was trained from scratch, without phenotype labels, on 169,120 profiles represented at 49,156 curated CpG sites, using the large public methylation resource and CpG panel assembled for MethylGPT [14]. Each token combines CpG identity with its quantitative methylation value. A Transformer encoder integrates these measurements, while genomic-rank rotary position encoding preserves chromosomal order and a dedicated CLS token provides a global representation for downstream adaptation [15].

To learn this representation, we adapt the whole cell expression decoder objective introduced in BMFM-RNA [16] to methylation as the Whole-CpG Encoder–Decoder (WCED). Two independently sampled partial views are generated from each methylation profile and processed by a shared encoder. For each view, a decoder predicts the measured profile from the CLS bottleneck, but the reconstruction loss is evaluated only at valid CpGs excluded from that encoder input, so the objective cannot be satisfied by copying the input forward. A complementary response to the same problem is offered by scConcept, which discards feature reconstruction altogether and pretrains cell

representations purely by contrasting disjoint gene-panel views [17]. We adopt that diagnosis but not that remedy: rather than replacing reconstruction, we retain it in bottleneck form and add an InfoNCE objective that aligns CLS representations of the two views from the same profile while distinguishing representations from different profiles [18]. WCED therefore applies two distinct pressures to the same embedding, which must both support recovery of unobserved methylation values and remain stable across CpG subsets.

We show that independently sampled CpG views preserve sample-specific identity and that the CLS bottleneck supports accurate reconstruction of CpGs withheld from the encoder. Without phenotype supervision, the pretrained representation captures substantial age- and tissue-associated variation, and fine-tuning for chronological-age prediction outperforms both a matched MethylGPT pipeline and an ElasticNet baseline on a common held-out cohort. At the same time, frozen representations encode strong study-specific structure, and transfer of age information to entirely unseen studies is heterogeneous. Together, these results establish MethylLlama as a reusable DNA methylation foundation model while identifying cohort heterogeneity as a central challenge for cross-study generalization.

2 Results

2.1 Large-scale self-supervised pretraining of MethylLlama

Learning a reusable representation of the methylome requires a training signal that does not depend on phenotype labels, because such labels are available for only a small and inconsistently annotated fraction of public methylation profiles. We therefore pretrained MethylLlama on 169,120 human DNA methylation profiles represented in a shared vocabulary of 49,156 CpG sites, without using chronological age, tissue, sex, dataset, or disease annotations at any point in the objective (Fig. 1a). The corpus was partitioned 80/10/10 by a fixed permutation into 135,296 training, 16,912 validation, and 16,912 held-out profiles; the held-out partition was never seen during pretraining and supports the reconstruction analyses below.

Each profile generates two independently sampled views, each containing 50% of its measured CpGs. A shared encoder compresses each view into a 256-dimensional CLS representation, from which a decoder predicts measured CpGs withheld from that view; a symmetric cross-view contrastive term aligns the two CLS representations of one profile while separating representations of different profiles (Fig. 1b–d). Because the two views are drawn independently rather than as a partition, their expected overlap is half of the sites in either view—equivalently, one quarter of all measured CpGs—a property we return to below. The encoder architecture and token construction are given in Methods and illustrated in Supplementary Fig. S1.

Downstream evaluation used the AltuMAge-derived collection of 10,988 profiles measured at 21,368 CpGs. Two distinct populations were derived from it, and the distinction governs how every subsequent result should be read. Frozen-representation diagnostics use the original train/validation/test partition (7,416/1,308/2,264 profiles). The supervised age benchmark applies

a prespecified age filter and methylation-fingerprint deduplication, removing 328 age outliers and 71 duplicates to leave 10,589 profiles, from which five stratified training/validation folds (7,177/1,263) and one fixed test set of 2,149 unique samples were constructed. Every fine-tuning and paired-comparison metric reported below uses that same 2,149-sample test set, for both model families and all five folds. Figure 1e summarizes the two downstream evaluation modes considered here: supervised age fine-tuning and frozen representation analysis.

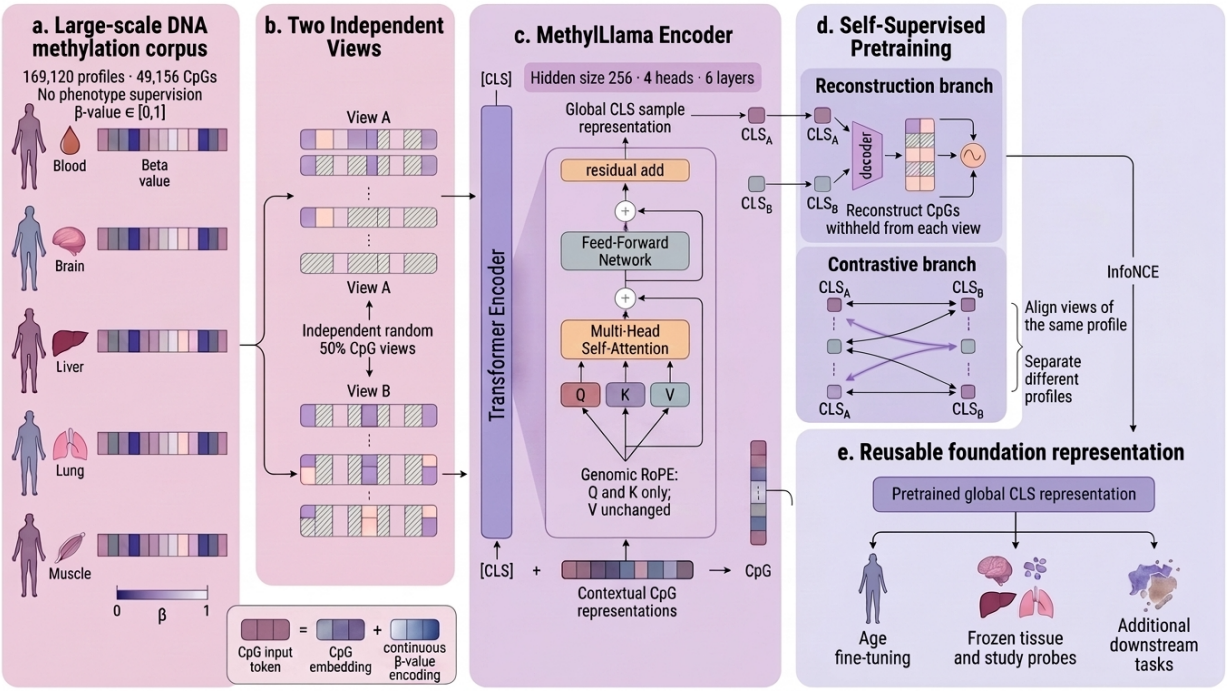


Figure 1: Overview of MethylLlama pretraining and downstream reuse. a, The self-supervised pretraining corpus comprises 169,120 human DNA methylation profiles represented at 49,156 selected CpG sites; phenotype annotations do not enter the objective. **b**, Each profile is sampled into two independent random views, each retaining 50% of its measured CpGs. Coloured cells denote CpGs present in a view and hatched cells denote CpGs omitted from that view. **c**, A shared six-layer Transformer encoder combines CpG identity with continuous methylation-value encoding and summarizes each view in a global CLS representation. Genomic-rank rotary position encoding is applied to the query and key projections within self-attention, while the value projection is unchanged. **d**, WCED couples reconstruction and cross-view contrast: a shared decoder uses each CLS representation to predict measured CpGs withheld from the corresponding input, while a symmetric InfoNCE objective aligns the two views of one profile against representations of other profiles. **e**, The pretrained global representation is reused for supervised chronological-age fine-tuning and frozen tissue and study probes, with further downstream tasks left for future evaluation. The loss is $\mathcal{L} = \mathcal{L}_{\text{recon}} + 0.05\,\mathcal{L}_{\text{InfoNCE}}$; additional architectural detail is provided in Supplementary Fig. S1.

2.2 WCED learns sample-specific representations from partial methylation profiles

If the CLS bottleneck is to serve as a reusable sample representation, it must satisfy two requirements that pull in opposite directions: it must be stable when a different subset of the methylome is observed, and it must remain specific enough to distinguish one profile from another. We tested

both directly.

Across five independent random draws of two half-profile views for each of 2,000 profiles, matched views had a mean cosine similarity of 0.984 ± 0.0004 against 0.499 ± 0.0002 for unmatched profiles, and the correct partner view was retrieved as the nearest neighbour for 73.8% of queries (range 73.1–74.5% across draws; chance 0.05%), rising to 96.1% within the ten nearest candidates (Fig. 2a,b). The representation is therefore stable under random 50% CpG subsampling while remaining specific to the individual profile.

Part of that stability is supplied by shared input rather than by the representation itself. Repeating the evaluation with strictly disjoint views—a random partition in which the two halves share no site—left mean matched similarity almost unchanged (0.971 ± 0.0005 versus 0.984) but reduced top-1 retrieval to 25.3% (range 24.6–26.0%). The dissociation is informative: mean similarity is insensitive to the change, whereas identifying the correct partner among 1,999 competitors depends on narrow margins against the most similar profiles, which the small loss of positive similarity is sufficient to erase. Cross-subset identity information is clearly retained—disjoint-view retrieval remains roughly 500-fold above chance and reaches 75.1% within ten candidates—but it is weaker than the overlapping-view figure alone implies. The encoder was trained with overlapping views, so the disjoint condition is out of distribution for it and these values are a conservative bound rather than an estimate of what the architecture achieves when trained for that regime (Supplementary Fig. S2).

A related question is whether a profile assayed at fewer sites embeds where its complete profile does. The embedding of a random 50% CpG subset had mean cosine similarity 0.991 to the embedding of its own complete profile (shuffled-pairing control, 0.496), and a ridge age probe fitted on complete-profile embeddings and applied without adaptation to subset embeddings of held-out profiles retained $R^2 = 0.835$ against $R^2 = 0.863$ on complete profiles (paired difference 0.028, 95% CI 0.014–0.044; Supplementary Fig. S3). These are random CpG subsets rather than the probe set of a particular array generation, so the result indicates robustness to which sites are observed and does not by itself establish transfer across measurement platforms.

We then asked whether the CLS bottleneck carries enough information to recover methylation values the encoder never saw. Reconstruction was evaluated on 5,000 profiles from the held-out partition of the pretraining corpus, encoding each from a 50% subset of its measured CpGs and scoring predictions only at measured CpGs absent from that input (1.12×10^8 evaluated positions). Predicted and observed beta values were closely correlated (Pearson $r = 0.972$, computed on a random sample of approximately 5.0×10^5 withheld positions), and the per-profile standardized reconstruction loss, computed over all withheld positions, was 0.0525 (Fig. 2c).

To test whether the decoder reads the sample representation rather than exploiting population-level regularities, we compared three controls on the same profiles under identical withheld masks. Mean squared error at withheld CpGs was 0.0056 for the model, against 0.0306 for a per-CpG mean predictor, 0.0505 for a decoder given standard normal vectors, and 0.0552 for a decoder given CLS embeddings shuffled across the batch. The paired per-profile difference against every

control excluded zero (Fig. 2d), and the model reduced error 5.4-fold relative to the per-CpG mean even though that baseline was computed on the evaluation cohort itself and is therefore advantaged. Shuffled CLS embeddings produced slightly worse reconstruction than random vectors, indicating that a real but mismatched sample representation is more misleading than no sample information at all. Together with cross-view retrieval, this establishes that the CLS bottleneck carries sample-specific information sufficient both to identify a profile and to predict methylation values never supplied to the encoder.

2.3 Self-supervised pretraining captures biologically informative variation

Pretraining optimized reconstruction and cross-view agreement, neither of which references biology. We therefore asked what biological information the frozen representation nonetheless contains, using linear probes fitted on the original AltuMAge partition (8,724 profiles for fitting, 2,264 for evaluation) with no update to the encoder.

A ridge regressor on frozen CLS embeddings predicted chronological age with $R^2 = 0.866$ and a median absolute error of 5.17 years. Applying the same probe to mean-pooled contextual CpG embeddings from the same model reduced performance to $R^2 = 0.795$ and MedAE 6.53 years (Fig. 4a). The CLS space also had a higher effective rank than the mean-pooled space and lower mean cosine anisotropy (Supplementary Table S2). The explicit sample token therefore retains broader and more age-informative variation than uniform pooling over CpG tokens, despite receiving no age supervision.

Biological information was not restricted to age: a linear classifier on frozen CLS embeddings attained balanced accuracy 0.936 across 37 sufficiently represented tissue classes (2,197 evaluation profiles; chance 0.027). These probes establish that biologically associated structure is linearly accessible in the pretrained representation under an in-distribution sample split. They do not show that biology has been separated from study or platform effects, because profiles from the same study can fall on both sides of this split—a question we address below.

2.4 MethylLlama improves age prediction under a matched benchmark

We next asked whether MethylLlama supports accurate chronological-age prediction under a matched benchmark. Five models were fine-tuned on different stratified training/validation folds and every selected checkpoint was evaluated on the same fixed 2,149-sample test cohort (age range 0–103 years; mean 45.53 years). Across folds, mean MedAE was 3.169 ± 0.040 years, mean MAE 4.493 ± 0.059 years, and mean R^2 0.9316 ± 0.0024 (mean $\pm$ sample SD; per-fold values in Supplementary Table S1). Because all five models were evaluated on the same samples, this dispersion measures sensitivity to the supervised training partition and is not a confidence interval for population performance (Fig. 3a). Averaging the five predictions per sample gave the primary MethylLlama estimate: MedAE 3.112 years, MAE 4.333 years, $R^2 = 0.9365$. MethylGPT was evaluated using the same fold assignments, held-out subjects, and fold-ensemble aggregation.

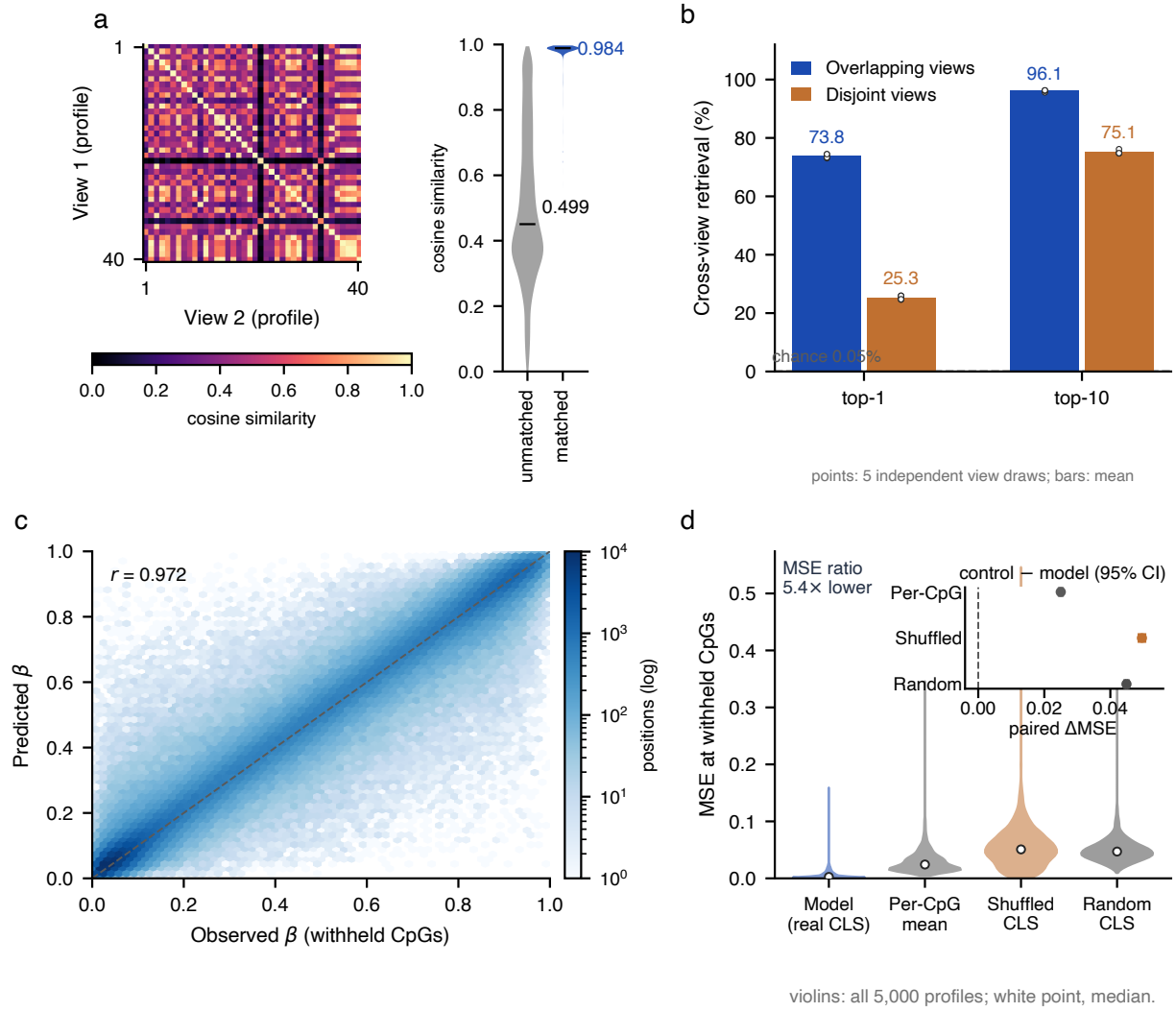


Figure 2: WCED learns view-stable, sample-specific representations that reconstruct withheld CpGs. **a**, Cosine similarity between CLS embeddings of two independently sampled 50%-CpG views. Heatmap: 40 profiles drawn at random under a fixed seed; the bright diagonal marks matched views of the same profile, and the two dark rows are individual profiles with genuinely low similarity to the rest of the draw rather than missing data. Violins: all 2,000 profiles of one draw, means annotated. **b**, Cross-view retrieval accuracy against the number of candidates considered, for overlapping and disjoint view constructions, with chance shown. Points give the mean across five independent view draws and vertical lines their range; top-1 and top-10 are annotated. **c**, Observed versus predicted β at CpGs withheld from the encoder input, for 5,000 profiles of the held-out pretraining partition (hexagonal binning, logarithmic colour scale; dashed line, identity). The panel displays a random sample of approximately 5.0×10^5 of the 1.12×10^8 evaluated positions, and the reported $r = 0.972$ was computed on that sample; reconstruction-error summaries in **d** were computed over all withheld positions. **d**, Per-profile mean squared error at withheld CpGs for the model and three controls evaluated on the same 5,000 profiles under identical withheld masks (violins, all profiles; white point, median). Inset: paired control-minus-model differences with 95% bootstrap confidence intervals over 10,000 resamples of profiles; the statistical unit is the profile.

The comparison used identical held-out samples, age labels, and fold assignments for both model families, and uncertainty was estimated by resampling the same test samples for both models. Throughout the text, table and figure, differences are signed so that positive values favour MethylLlama: error differences are comparator minus MethylLlama, and R^2 differences are MethylLlama minus comparator. MethylLlama reduced MedAE by 0.589 years (95% CI 0.338–0.815) and MAE by 0.773 years (0.583–0.968) relative to MethylGPT, and increased R^2 by 0.0254 (0.0183–0.0329); all three intervals excluded zero (Table 1, Fig. 3b,c,d).

We additionally evaluated a conventional linear reference using ElasticNetCV [19], with hyperparameters selected by internal cross-validation on the pooled training and validation profiles and the common test set withheld until final evaluation. The resulting model used 1,036 non-zero coefficients among 21,368 CpGs and achieved MedAE 3.437 years, MAE 4.731 years, and $R^2 = 0.9322$. The MethylLlama ensemble had better point estimates than both comparators on this test set. Because ElasticNet was evaluated as a single fitted model whose sample-level predictions were not retained, that comparison is descriptive and carries no paired confidence interval.

The benchmark is matched at the sample and split levels, but the two pipelines differ in their treatment of 1,760 CpGs outside the shared pretraining reference: MethylLlama retains their measured values under a common unknown-CpG identity, whereas MethylGPT excludes them. The result therefore establishes a predictive advantage for the implemented MethylLlama pipeline as a whole, and does not isolate the contribution of the WCED objective; an ablation that varies the pretraining objective while holding the pipeline fixed would be required for that attribution.

Table 1: Age prediction on the common 2,149-sample held-out test set. The first three rows are test-set point estimates. MethylLlama and MethylGPT are five-fold prediction ensembles; ElasticNetCV is a single model selected by internal cross-validation. The MethylGPT comparison row reports paired improvement of MethylLlama with 95% bootstrap confidence intervals over 10,000 resamples of the test samples, signed so that positive values favour MethylLlama for all three endpoints (Δ error = comparator − MethylLlama; ΔR^2 = MethylLlama − comparator). ElasticNet sample-level predictions were not retained, so its improvement row is descriptive and has no paired interval.

Model or comparison	MedAE (years)	MAE (years)	R^2
MethylLlama (five-fold ensemble)	3.112	4.333	0.9365
ElasticNetCV (single model)	3.437	4.731	0.9322
MethylGPT (five-fold ensemble)	3.701	5.105	0.9110
Improvement over MethylGPT (paired)	0.589 [0.338, 0.815]	0.773 [0.583, 0.968]	0.0254 [0.0183, 0.0329]
Improvement over ElasticNetCV (descriptive)	0.325	0.398	0.0043

2.5 Pretrained representations retain biological and study-associated structure

The probes above were fitted and evaluated on a split that mixes studies, so they cannot distinguish biological signal from cohort signal. We therefore asked how much study-associated structure the frozen representation carries, and whether age information transfers to a study withheld entirely from probe fitting.

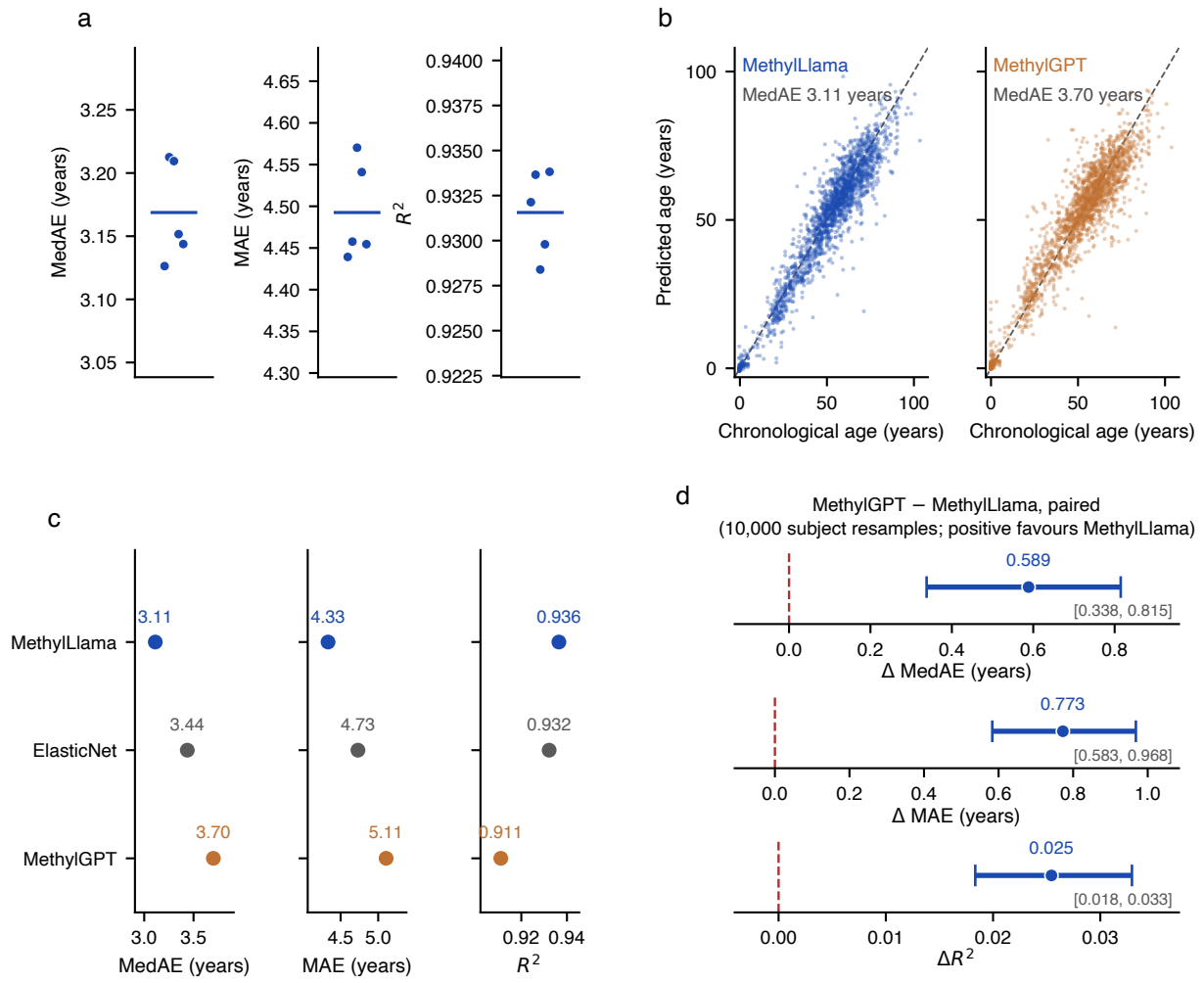


Figure 3: Age prediction is stable across supervised partitions and improves on matched neural and linear references. a, Stability across five supervised training partitions. Each point is one fine-tuned model evaluated on the same fixed 2,149-sample test cohort; horizontal lines are the means across partitions. The spread reflects sensitivity to the training partition, not population uncertainty, because the test samples are shared. **b**, Ensemble-predicted versus chronological age for MethylLlama and MethylGPT on the same 2,149 samples, on identical axes and scales; each point is one sample and dashed lines indicate identity. **c**, Ensemble point estimates for MethylLlama and MethylGPT with the single-model ElasticNetCV reference. **d**, Paired differences against MethylLlama with 95% bootstrap confidence intervals over 10,000 resamples of the test samples, signed so that positive values favour MethylLlama for all three endpoints; the vertical dashed line marks no difference. Statistical unit is the test sample; $n = 2{,}149$ throughout.

Both tissue and dataset identity were strongly linearly decodable from frozen CLS embeddings: balanced accuracy 0.936 across 37 tissue classes (chance 0.027) and 0.956 across 85 datasets (chance 0.012) (Fig. 4b). These two values should not be compared directly, because the tasks differ in class number and in evaluation size (2,197 versus 2,006 profiles) and both are single fixed-split point estimates. The interpretable finding is that the representation simultaneously retains substantial biological and substantial cohort-specific structure. Tissue and dataset labels are themselves partially confounded in public methylation collections, so the tissue-probe accuracy cannot be read as a study-independent biological measure.

Cross-study transfer makes the consequence explicit. A ridge age probe fitted on a random 80/20 split of the same embeddings achieved $R^2 = 0.856$ (MedAE 5.05 years, 2,198 evaluation profiles), whereas the median leave-one-dataset-out R^2 across 11 eligible studies was 0.261 with a median MedAE of 8.32 years, and three studies yielded negative R^2 (Fig. 4c). Readily decodable age information within a mixed-study split therefore does not ensure transfer to an unseen cohort.

We did not detect an association between transfer success and either obvious cohort descriptor: age dispersion (Spearman $\rho = +0.28$, $P = 0.40$) or sample size ($\rho = +0.26$, $P = 0.43$). This study-level analysis contains only 11 cohorts and is underpowered to detect modest associations, so these are non-detections rather than evidence of no relationship. The failing and succeeding cohorts nonetheless overlap on both axes—one cohort of 118 profiles with an age SD of 10.5 years transferred at $R^2 = 0.42$, whereas a larger cohort of 282 profiles with a wider age SD of 13.9 years transferred at $R^2 = -4.63$—which is consistent with study-specific biological or technical structure, although the responsible factors were not isolated here.

This is a diagnostic of the frozen representation and not an external validation of the fine-tuned age model, which was neither trained nor evaluated under a study-held-out design. Tissue, disease status, platform, and age distribution also remain entangled with study identity. Establishing study-independent age prediction will require grouped study-held-out fine-tuning with cohort-level uncertainty; the present analysis identifies cross-study transfer as an unresolved limitation.

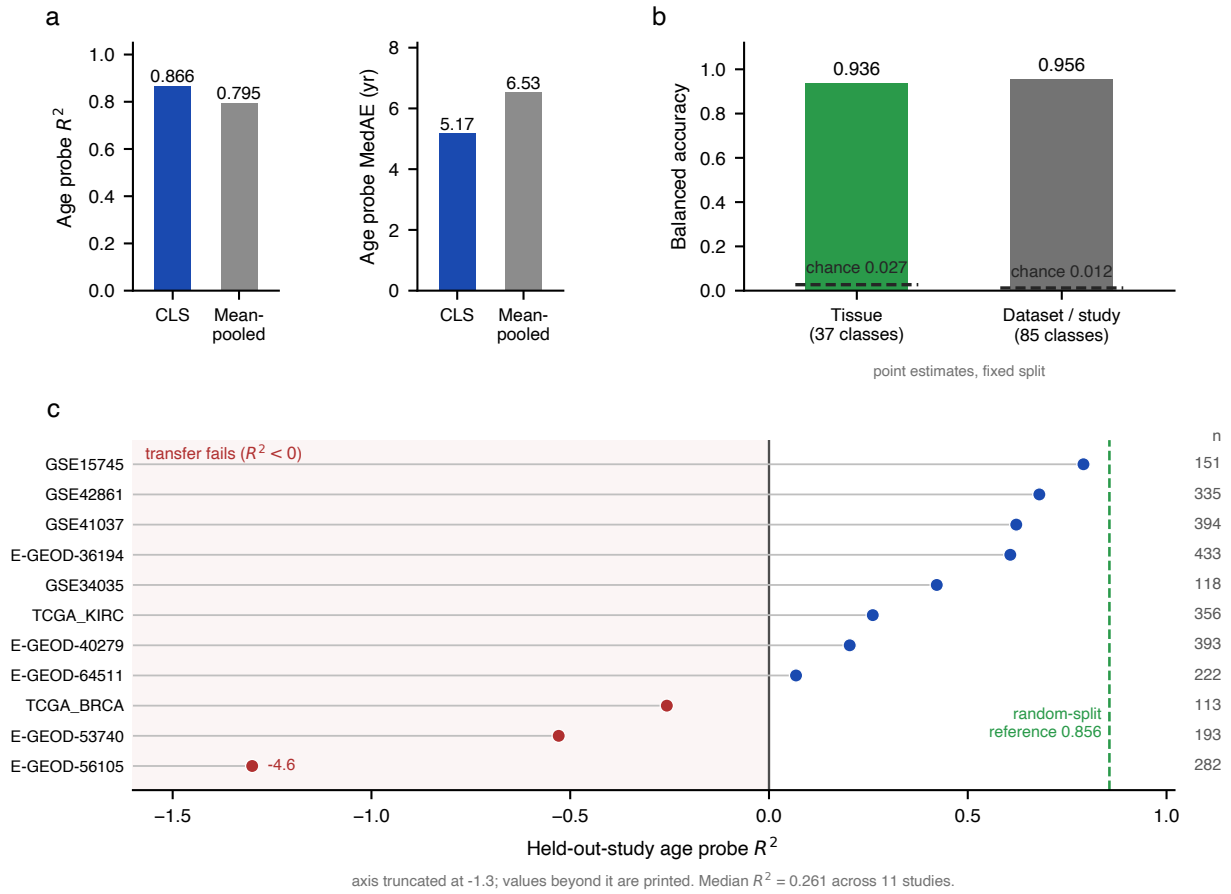


Figure 4: Biological structure, study structure, and the limits of cross-study transfer. **a**, Frozen linear age-probe performance using the pretrained CLS representation or mean-pooled contextual CpG embeddings, fitted on 8,724 and evaluated on 2,264 profiles of the original AltuMAge partition. **b**, Balanced accuracy of frozen CLS linear probes for tissue (37 classes, 2,197 profiles) and dataset identity (85 classes, 2,006 profiles); dashed segments mark the corresponding chance levels. The two tasks differ in class number and evaluation size and are not directly comparable to one another. **c**, Leave-one-dataset-out age-probe R^2 for the 11 studies meeting the prespecified size ($n \geq 100$) and age-dispersion (SD ≥ 10 years) thresholds, ordered by R^2; each row is one held-out study, with its cohort size given alongside. The dashed vertical line is the in-distribution random-split reference ($R^2 = 0.856$) and the solid line marks $R^2 = 0$; studies with negative R^2 are shown in the shaded region. The axis is truncated at $R^2 = -1.3$ and the more extreme value is printed at the boundary. All panels report fixed-split point estimates from a single frozen encoder; these analyses diagnose the pretrained representation and do not constitute external validation of the fine-tuned model.

3 Discussion

We developed a partial-view self-supervised framework for learning global representations of DNA methylation samples. Unlike objectives that operate only at the level of masked features, WCED routes reconstruction through a CLS bottleneck and evaluates reconstruction at CpGs not provided to the encoder. Contrastive view matching directly trains the same sample embedding that is subsequently transferred to downstream analysis. The combination targets a distinction that is easy to miss in high-dimensional molecular modeling: successful prediction of masked features is not equivalent to learning a useful global representation.

The current results establish three narrow but meaningful points. First, the encoder can compress a partial methylation profile while retaining information about unobserved sites. Second, independently sampled views of the same profile remain identifiable in representation space. Third, the resulting encoder can be adapted reproducibly to age prediction. The CLS representation contained age-associated information before fine-tuning and retained more usable variation than mean pooling, consistent with the intended sample-level bottleneck.

The paired benchmark establishes better predictive performance for the implemented MethylLlama pipeline than for MethylGPT on the matched held-out samples. It does not, however, isolate the source of that difference: the pipelines vary in architecture, optimization, and their handling of 1,760 CpGs outside the shared pretraining reference. ElasticNet provided a strong non-neural reference, with performance close to the MethylLlama ensemble. Although all three point estimates favored MethylLlama, sample-level ElasticNet predictions were not retained, so statistical uncertainty is currently available only for the paired MethylGPT comparison. Matched objective and initialization ablations are still required before attributing the downstream gain specifically to WCED or to self-supervised pretraining.

The strongest limitation is cohort structure. Dataset identity is highly recoverable from the pretrained embeddings, showing that study- or batch-specific signals remain prominent. Random sample partitions can therefore overestimate transfer when samples from the same source study appear in model development and evaluation. This issue also changes how representation visualizations should be read. Separation by tissue may reflect genuine biology, but clean clusters can coexist with strong study effects. Quantitative study-held-out tests should carry more evidential weight than visually distinct UMAP clusters.

The representation analyses are therefore best viewed as diagnostics. Attention concentration, genomic locality, and CpG embedding structure can motivate hypotheses, but they do not by themselves identify causal age-associated loci or establish biological mechanism. Stronger interpretation would require stability across folds and seeds, perturbation-based importance, enrichment against a prespecified CpG background, and correction for multiple testing.

Further limitations include reliance on array-selected CpGs, incomplete evaluation across platforms, a single firmly established downstream phenotype, and no prospective clinical validation. A broad foundation-model claim would require transfer across multiple tasks and cohorts. At the present stage, “self-supervised methylation encoder” or “methylation representation model” more

precisely describes the demonstrated scope.

In summary, the evidence supports WCED as a method for learning partial-view-consistent global methylation representations and supports MethylLlama as a stable age-prediction encoder on the held-out evaluation set. Whether the objective improves upon matched alternatives and whether the representation transfers to unseen studies remain the decisive experiments for the final manuscript.

4 Methods

4.1 Datasets and study design

Self-supervised pretraining used 169,120 public methylation profiles represented over 49,156 CpG sites. Profiles were partitioned once into training, validation, and held-out portions (80/10/10) by a fixed random permutation (seed 42); the held-out portion of 16,912 profiles was never seen during pretraining and provides the evaluation set for the reconstruction analyses reported below. The downstream age corpus was derived from the AltuMAge collection [20] and comprised 10,988 profiles over 21,368 CpG sites, spanning 37 tissue categories and 85 source studies. After removing 328 profiles with implausible age annotations (< 0 or > 120 years) and 71 duplicate profiles, 10,589 remained: 7,177 training, 1,263 validation, and a common held-out test set of 2,149 unique samples. Because the same test samples were evaluated by all five fold-trained models, variation across models is reported as training-fold stability rather than as an independent population confidence interval.

Split integrity. Partitions were defined at the level of individual profiles, indexed by their public sample accession; donor-level grouping was not available for the majority of source studies, which we note as a limitation. Duplicate and technical-replicate profiles were identified by methylation fingerprinting across all partitions, yielding 77 duplicate pairs spanning 136 profiles; within each connected component of the resulting graph the lexicographically first accession was retained and the remaining 71 profiles were excluded. The same exclusion list and age filter were applied identically to every model and baseline evaluated here, and set equality of the resulting 2,149 test accessions was verified programmatically rather than inferred from matching counts. Exact split manifests and their hashes will accompany the release.

Preprocessing. Beta values were restricted to $[0, 1]$. Pretraining used the full 49,156-CpG reference panel; downstream modelling used the 21,368 CpGs measured in the AltuMAge collection. Missing measurements were excluded from view sampling and from all loss terms through an explicit validity mask, so that no imputed value ever contributed to training or evaluation. CpGs were ordered by chromosome and genomic coordinate on the hg38 assembly using the Illumina HM450 manifest. No target-age labels were used during self-supervised pretraining: although the implementation exposes an optional auxiliary age-regression term, its weight was set to zero for the pretraining run reported here.

4.2 Input representation and encoder

Each observed CpG was represented by its site identity c_i and beta value b_i. The token embedding was

$$\boldsymbol{x}_i = s_c\, \boldsymbol{e}_{c_i} + f_\beta(b_i), \tag{1}$$

where $\boldsymbol{e}_{c_i}$ is a learned CpG embedding, s_c is a learned or initialized scale, and f_β is a continuous-value encoder based on sinusoidal basis functions followed by a linear projection. A CLS token was prepended to every partial profile.

The encoder used six transformer layers with hidden size 256, four attention heads (head dimension 64), feed-forward width 512, and a vocabulary of 49,161 entries (49,156 CpG sites plus five special tokens). Transformer blocks used pre-normalization with RMSNorm [21], multi-head self-attention [22], SwiGLU feed-forward layers, and rotary position embeddings [15, 23]. RoPE position identifiers were assigned using genomic rank rather than arbitrary input order. The pooled CLS output served as the sample representation.

4.3 Partial-view WCED pretraining

WCED adapts the Whole-Cell Expression Decoder introduced for transcriptomic foundation models [16] to array-derived DNA methylation. The original objective reconstructs a whole-cell expression profile through a CLS bottleneck; our formulation reconstructs CpG methylation values and evaluates reconstruction only at sites excluded from the encoder input.

The paired-view design was inspired by scConcept, which contrasts representations of different gene-panel views of the same cell [17]. For each methylation profile, two independently sampled views $V_i^{(1)}$ and $V_i^{(2)}$ contained 50% of valid CpGs. A shared encoder E_θ produced sample representations $\boldsymbol{h}_i^{(v)} = E_\theta(V_i^{(v)})_{\mathrm{CLS}}$. A shallow decoder D_ϕ mapped each CLS representation to a full-vector prediction $\hat{\boldsymbol{b}}_i^{(v)}$.

Reconstruction error was evaluated only over valid CpGs absent from the corresponding input view, denoted $H_i^{(v)}$:

$$\mathcal{L}_{\mathrm{recon}} = \frac{1}{2}\sum_{v=1}^{2} \frac{1}{|H_i^{(v)}|} \sum_{j \in H_i^{(v)}} \left(\tilde{b}_{ij} - \tilde{\hat{b}}_{ij}^{(v)} \right)^2, \tag{2}$$

where tildes denote per-sample standardisation: within each view, predicted and observed values at the withheld positions are centred and scaled to unit variance before the squared error is taken. This prevents the decoder from reducing the loss by predicting each CpG's marginal mean, and it makes the reported reconstruction loss scale-free.

A two-layer projection head g ($256 \to 128 \to 128$, ReLU) mapped CLS embeddings to L_2-normalized vectors $\boldsymbol{z}_i^{(v)}$. The symmetric cross-view InfoNCE loss used matched views as positives and other samples in the batch as negatives [18]:

$$\mathcal{L}_{\mathrm{ctr}} = -\frac{1}{2N}\sum_{i=1}^{N} \left[\log \frac{\exp(\boldsymbol{z}_i^{(1)\top}\boldsymbol{z}_i^{(2)}/\tau)}{\sum_j \exp(\boldsymbol{z}_i^{(1)\top}\boldsymbol{z}_j^{(2)}/\tau)} + \log \frac{\exp(\boldsymbol{z}_i^{(2)\top}\boldsymbol{z}_i^{(1)}/\tau)}{\sum_j \exp(\boldsymbol{z}_i^{(2)\top}\boldsymbol{z}_j^{(1)}/\tau)} \right]. \tag{3}$$

The complete objective was $\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda \mathcal{L}_{\text{ctr}}$, with a fixed temperature $\tau = 0.1$ and $\lambda = 0.05$.

Relationship to contrastive single-cell pretraining. The contrastive component follows the cell-view formulation of scConcept [17], and we state the differences explicitly because they bound interpretation. First, and most importantly, contrastive alignment is the *sole* pretraining objective in that framework, which dispenses with reconstruction altogether; here it is an auxiliary term ($\lambda = 0.05$) accompanying withheld-CpG reconstruction. We adopt the underlying diagnosis—that feature-level reconstruction does not by itself optimise the sample-level embedding—but address it with two complementary mechanisms: routing reconstruction through the CLS bottleneck and scoring it only at withheld sites, and adding explicit cross-view alignment pressure on the same representation. Second, their two views are disjoint gene panels, whereas our two 50% CpG subsets are drawn independently and may overlap, which makes view alignment a comparatively easier task. Third, their negative set additionally contains embeddings from the anchor's own view, which discourages the encoder from representing panel identity rather than sample identity; our negative set contains only cross-view embeddings. Fourth, their temperature is learned during training whereas ours is fixed, and their contrastive loss is computed on the CLS embedding directly rather than through a projection head. Finally, their mini-batches are drawn from a single source dataset so that discrimination cannot be solved using inter-study technical differences, whereas ours are sampled at random from the pretraining corpus; we return to this point in the Discussion.

4.4 Age fine-tuning and comparators

A two-layer regression head ($256 \to 128 \to 1$, dropout 0.1) was applied to the CLS representation. The encoder was frozen for the first ten epochs while the head adapted, and then unfrozen and optimised jointly at a reduced encoder learning rate (2×10^{-5} versus 10^{-4} for the head). Fine-tuning used the complete measured profile as input rather than a partial view, and was optimised with a Huber loss on standardised age. Model selection used validation MedAE without access to the held-out test labels.

MethylGPT [14] was evaluated using the same source dataset, age labels, five fold assignments, and 2,149 held-out sample identifiers. Five predictions per sample were averaged within each model family before comparison. The pipelines differed in feature handling: 19,608 of the 21,368 downstream CpGs occurred in the common pretraining reference, whereas the remaining 1,760 CpGs were retained by MethylLlama under a shared unknown-site identity and excluded by the MethylGPT conversion pipeline.

As a non-neural reference, we fitted ElasticNetCV [19] to all 21,368 beta-value features after standardization based only on the pooled training and validation profiles. The documented l_1-ratio grid (0.1, 0.5, 0.7, 0.9, 0.95, 0.99, and 1.0), an automatically derived 100-value alpha path, and five-fold internal cross-validation were used once without post-hoc expansion of the search space. The selected model had $\alpha = 0.10883$, l_1 ratio 0.7, and 1,036 non-zero coefficients. The common test set was not used for hyperparameter selection. A single-fold random-initialization experiment was

retained as exploratory and was not used for the primary comparison; matched objective ablations remain necessary to attribute downstream performance specifically to WCED.

4.5 Evaluation and statistical analysis

Primary endpoints were median absolute error (MedAE) and mean absolute error (MAE), measured in years. R^2 was secondary. Fold-level means and sample standard deviations quantify sensitivity to the training fold and are not population confidence intervals. For the MethylLlama–MethylGPT comparison, 95% confidence intervals were obtained from 10,000 paired percentile-bootstrap replicates with random seed 0. Held-out samples, rather than fold-level predictions, were resampled; the same bootstrap indices were applied to both model ensembles. The ElasticNet result is reported as a point-estimate reference because per-sample predictions were not retained for a paired analysis.

Representation analyses included matched-versus-unmatched view cosine similarity, cross-view retrieval, linear and nonlinear age probes, effective rank, and CLS-versus-mean pooling. All probe splits were isolated from probe fitting and hyperparameter selection.

Withheld-CpG reconstruction. Reconstruction was evaluated on 5,000 profiles drawn from the held-out portion of the pretraining corpus, which was excluded from pretraining by the fixed partition described above. Each profile was encoded from a 50% subset of its measured CpGs, and predictions were scored only at measured CpGs absent from that input—the same mask used by the training objective—giving 1.12×10^8 evaluated positions. Three controls were computed on the identical profiles under the identical masks: a per-CpG mean predictor, the decoder applied to CLS embeddings shuffled across the batch, and the decoder applied to standard normal vectors. The per-CpG means were computed on the evaluation cohort itself, which favours that baseline and therefore makes the comparison conservative.

Cross-view consistency. Two 50% views were encoded for each of 2,000 profiles and compared by cosine similarity of the resulting CLS embeddings. We report the mean similarity for matched and unmatched pairs, and retrieval accuracy at rank k: the fraction of first views whose matched second view falls within the k nearest of all 2,000 candidates. Retrieval at $k = 1$ is the strictest point on this curve; chance is $k/2000$. The whole evaluation was repeated for five independent random view draws, and we report the mean and range across draws rather than a single draw.

Two view constructions were compared. In the first, matching the construction used during pretraining, the two views are drawn independently and therefore share approximately half of their CpGs in expectation. In the second, they form a random partition of the valid CpGs and share no site, following the disjoint-panel construction of Bahrami et al. [17]. The disjoint condition is out of distribution for an encoder trained under the overlapping construction, and we therefore interpret it as a conservative bound on cross-subset consistency rather than as a matched comparison. Both conditions used the same profiles, checkpoint, and encoder call, so they differ only in how the two views are formed.

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